

Q15-general: Heisenberg-highest structure of $q_e^{(n)}$ in $\widehat{\mathfrak{sl}}_e$ -Fock

Clio

2026-08-10

Abstract

Let $e \geq 2$, $k' \geq 1$, and $n = ek'$. Let $\mathcal{F}$ denote the level-1 Fock space of $\widehat{\mathfrak{sl}}_e$, with standard basis $\{|\lambda\rangle : \lambda \text{ a partition}\}$ and Misra–Miwa crystal $(\tilde{e}_i, \tilde{f}_i)_{i \in \mathbb{Z}/e\mathbb{Z}}$. Set

$$v_{k',e} := P_e^{k'} |\emptyset\rangle = \sum_{\lambda \vdash n} \chi_{(ek')}^\lambda |\lambda\rangle,$$

where P_e is the classical Heisenberg operator $p_e \cdot$ on $\mathcal{F}$.

We prove three structural properties of $v_{k',e}$:

- (a) $v_{k',e}$ is $\widehat{\mathfrak{sl}}_e$ -weight-homogeneous of weight $\Lambda_0 - k'\delta$.
- (b) $\tilde{e}_i^{k'+1}(v_{k',e}) = 0$ for every $i \in \{0, 1, \dots, e-1\}$.
- (c) Every length- n Kashiwara word $(w_1, \dots, w_n) \in \{0, \dots, e-1\}^n$ satisfying $\tilde{e}_{w_n} \cdots \tilde{e}_{w_1}(v_{k',e}) \in \mathbb{Q}^\times |\emptyset\rangle$ contains each letter exactly k' times.

The stretch statement from the PROVE seed — $\tilde{e}_i^{k'}(v_{k',e}) \neq 0$ — is *false*: we prove the strictly stronger vanishing $\tilde{e}_0^{k'}(v_{k',e}) = 0$ for every $e \geq 2, k' \geq 1$, and record a general weight-space vanishing bound (Proposition 9). In fact, computation at the four cases $(n, e, k') = (6, 3, 2), (8, 2, 4), (9, 3, 3), (12, 3, 4)$ shows the far stronger empirical fact $\varepsilon_i(v_{k',e}) = 1$ for every i , which we state as Conjecture 11 but do not prove here.

Every claim proved below is verified computationally; the log is included as §8.

1 Setup and conventions

Let $e \geq 2$. Denote by $\widehat{\mathfrak{sl}}_e$ the affine Kac–Moody algebra with simple roots $\alpha_0, \alpha_1, \dots, \alpha_{e-1}$, fundamental weight Λ_0 at level 1, and null root $\delta = \alpha_0 + \alpha_1 + \cdots + \alpha_{e-1}$.

Partitions are weakly decreasing tuples $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots)$ of positive integers; cells are pairs (r, c) with 0-indexed row and column, so that $\lambda = \{(r, c) : 0 \leq r, 0 \leq c < \lambda_{r+1}\}$. The residue of a cell is

$$\text{res}(r, c) := (c - r) \bmod e \in \{0, 1, \dots, e-1\},$$

and the residue count is

$$n_i(\lambda) := \#\{(r, c) \in \lambda : \text{res}(r, c) = i\}.$$

The level-1 Fock space $\mathcal{F}$ has $\mathbb{Q}$ -basis $\{|\lambda\rangle\}$ indexed by all partitions, with vacuum $|\emptyset\rangle$. The $\widehat{\mathfrak{sl}}_e$ -weight of a basis vector, modulo $\mathbb{Q}\delta$, is

$$\text{wt}(|\lambda\rangle) = \Lambda_0 - \sum_{i=0}^{e-1} n_i(\lambda) \alpha_i. \tag{1}$$

(All partitions of the same size share the same d -eigenvalue, so this fixes the affine weight class coherently on any weight-homogeneous, size-homogeneous subspace.)

The Misra–Miwa Kashiwara operators $\tilde{e}_i, \tilde{f}_i$ act on the basis via the good-removable and good-addable i -node rules (Ariki, Kleshchev). We use only two properties:

- (K1) $\tilde{e}_i$ is $\mathbb{Q}$ -linear on $\mathcal{F}$; on a basis vector $|\lambda\rangle$ it returns either 0 or $|\mu\rangle$ for a partition $\mu \subset \lambda$ with $|\mu| = |\lambda| - 1$, $n_i(\mu) = n_i(\lambda) - 1$, and $n_j(\mu) = n_j(\lambda)$ for $j \neq i$.
- (K2) Consequently $\tilde{e}_i$ shifts weight by $+\alpha_i$: for any weight-homogeneous $u \in \mathcal{F}$, $\text{wt}(\tilde{e}_i u) = \text{wt}(u) + \alpha_i$ whenever $\tilde{e}_i u \neq 0$.

The Heisenberg operator P_e acts on $\mathcal{F}$ by the signed ribbon-addition rule

$$P_e |\lambda\rangle = \sum_{\substack{\mu \supset \lambda \\ \mu/\lambda \text{ an } e\text{-ribbon}}} (-1)^{\text{ht}(\mu/\lambda)} |\mu\rangle,$$

where $\text{ht}(\mu/\lambda) = (\# \text{ rows spanned by the ribbon}) - 1$. Iterating and applying the ribbon Murnaghan–Nakayama rule yields

$$v_{k',e} := P_e^{k'} |\emptyset\rangle = \sum_{\lambda \vdash n} \chi_{(e^{k'})}^\lambda |\lambda\rangle, \quad n = ek'. \quad (2)$$

2 The ribbon residue lemma

Lemma 1 (Residues of an e -ribbon). *Let $\lambda \subsetneq \mu$ be partitions with $\mu \setminus \lambda$ an e -ribbon (a connected border strip of e cells). Then the residues of the cells of $\mu \setminus \lambda$ form a complete residue system modulo e : each residue in $\{0, 1, \dots, e-1\}$ appears exactly once.*

Proof. List the cells of $\mu \setminus \lambda$ from northeast to southwest as $(r_1, c_1), (r_2, c_2), \dots, (r_e, c_e)$. The ribbon condition forces each consecutive step to be either a left move $(r_{k+1}, c_{k+1}) = (r_k, c_k - 1)$ or a down move $(r_{k+1}, c_{k+1}) = (r_k + 1, c_k)$. In both cases

$$\text{res}(r_{k+1}, c_{k+1}) = (c_{k+1} - r_{k+1}) \bmod e = (c_k - r_k - 1) \bmod e = \text{res}(r_k, c_k) - 1 \pmod{e}.$$

Hence the residues of the e cells are $\text{res}(r_1, c_1), \text{res}(r_1, c_1) - 1, \dots, \text{res}(r_1, c_1) - (e-1)$ modulo e : a complete residue system. $\square$

Corollary 2. *If λ has empty e -core (equivalently, λ can be built from $\emptyset$ by adding $|\lambda|/e$ successive e -ribbons), then $n_i(\lambda) = |\lambda|/e$ for every $i \in \{0, 1, \dots, e-1\}$. In particular, when $|\lambda| = n = ek'$, every λ with empty e -core has residue count $(k', k', \dots, k')$.*

Proof. Immediate: each ribbon adds one cell of each residue by Lemma 1. $\square$

Remark 3 (Converse and support of $v_{k',e}$). *The ribbon Murnaghan–Nakayama rule gives $\chi_{(e^{k'})}^\lambda = \sum_T (-1)^{\text{ht}(T)}$, summed over e -ribbon tableaux T of shape λ (equivalently, sequences of length k' of e -ribbon additions building λ from $\emptyset$). No such tableau exists unless λ has empty e -core, so*

$$\chi_{(e^{k'})}^\lambda \neq 0 \implies \lambda \text{ has empty } e\text{-core}.$$

Combined with Corollary 2, every λ in the support of $v_{k',e}$ satisfies $n_i(\lambda) = k'$ for all i .

3 Weight-homogeneity

Theorem 4 (Part (a)). $v_{k',e}$ is $\widehat{\mathfrak{sl}}_e$ -weight-homogeneous of weight $\Lambda_0 - k'\delta$.

Proof. By Remark 3, every λ with $\chi_{(e^{k'})}^\lambda \neq 0$ satisfies $n_i(\lambda) = k'$ for all i . Substituting into (1),

$$\text{wt}(|\lambda\rangle) = \Lambda_0 - \sum_i k' \alpha_i = \Lambda_0 - k' \sum_i \alpha_i = \Lambda_0 - k' \delta.$$

Every basis element in the support of $v_{k',e}$ has the same weight $\Lambda_0 - k' \delta$. $\square$

4 Depth- $(k' + 1)$ Kashiwara killing

Theorem 5 (Part (b)). $\tilde{e}_i^{k'+1}(v_{k',e}) = 0$ for every $i \in \{0, 1, \dots, e-1\}$.

Proof. By (K2), $\tilde{e}_i$ shifts weight by $+\alpha_i$. If $\tilde{e}_i^{k'+1}(v_{k',e}) \neq 0$, it is a $\mathbb{Q}$ -linear combination of basis vectors $|\mu\rangle$ of weight

$$\Lambda_0 - k'\delta + (k' + 1)\alpha_i = \Lambda_0 - \sum_j (k' - (k' + 1)[j = i])\alpha_j.$$

Comparing with (1) using the linear independence of $\alpha_0, \dots, \alpha_{e-1}$ (they are simple roots of the affine Kac–Moody algebra), any such μ would satisfy $n_j(\mu) = k'$ for $j \neq i$ and $n_i(\mu) = -1$. Residue counts are non-negative, so no such μ exists. The weight space is 0, so $\tilde{e}_i^{k'+1}(v_{k',e}) = 0$. $\square$

5 Length- n paths to the vacuum have balanced letter counts

Theorem 6 (Part (c)). Let $w = (w_1, w_2, \dots, w_n) \in \{0, 1, \dots, e-1\}^n$ and set $m_i(w) := \#\{j : w_j = i\}$. If

$$\tilde{e}_{w_n} \tilde{e}_{w_{n-1}} \cdots \tilde{e}_{w_1}(v_{k',e}) \in \mathbb{Q}^\times |\emptyset\rangle,$$

then $m_i(w) = k'$ for every $i \in \{0, \dots, e-1\}$.

Proof. Applying the composite operator shifts weight by $\sum_j \alpha_{w_j} = \sum_i m_i(w) \alpha_i$. Applied to $v_{k',e}$ (weight $\Lambda_0 - k'\delta$ by Theorem 4), the result has weight

$$\Lambda_0 - k'\delta + \sum_i m_i(w) \alpha_i = \Lambda_0 - \sum_i (k' - m_i(w)) \alpha_i.$$

Setting this equal to $\text{wt}(|\emptyset\rangle) = \Lambda_0$ and using linear independence of the α_i forces $m_i(w) = k'$ for every i . $\square$

Corollary 7. The number of length- n words potentially inducing a nonzero path $v_{k',e} \rightarrow |\emptyset\rangle$ is at most $\binom{n}{k', k', \dots, k'} = n!/(k'!)^e$.

6 Correction of the seed's stretch conjecture

The PROVE seed proposed as a stretch conjecture that $\tilde{e}_i^{k'}(v_{k',e}) \neq 0$ for every i . This is *false*. The obstruction strengthens Theorem 5: the target weight space for $\tilde{e}_0^{k'}$ is empty in Fock.

Theorem 8 ($\tilde{e}_0$ -vanishing at depth k'). *For every $e \geq 2$ and $k' \geq 1$,*

$$\tilde{e}_0^{k'}(v_{k',e}) = 0,$$

and consequently $\varepsilon_0(v_{k',e}) := \max\{m : \tilde{e}_0^m(v_{k',e}) \neq 0\} \leq k' - 1$.

Proof. By the same weight-space argument as Theorem 5: any $|\mu\rangle$ in the support of $\tilde{e}_0^{k'}(v_{k',e})$ would have residue count $(0, k', k', \dots, k')$ and size $|\mu| = ek' - k' = (e-1)k'$. But $n_0(\mu) = 0$ forces $\mu = \emptyset$, since $(0,0)$ has $\text{res}(0,0) = 0$ and belongs to every nonempty partition. Then $|\mu| = 0$, contradicting $|\mu| = (e-1)k' > 0$ (which holds because $e \geq 2$ and $k' \geq 1$). The weight space vanishes, so $\tilde{e}_0^{k'}(v_{k',e}) = 0$. $\square$

Proposition 9 (General weight-space vanishing bound). *For every $i \in \{0, \dots, e-1\}$ and $m \geq 0$,*

$$\tilde{e}_i^m(v_{k',e}) = 0 \text{ whenever no partition } \mu \text{ satisfies } \begin{cases} |\mu| = ek' - m, \\ n_i(\mu) = k' - m, \\ n_j(\mu) = k' \text{ for } j \neq i. \end{cases}$$

Equivalently, $\varepsilon_i(v_{k',e}) \leq M_i(k', e)$ where $M_i(k', e)$ is the largest $m \geq 0$ for which such μ exists.

Proof. Identical to the proof of Theorem 5, replacing $k' + 1$ by m : the target weight space is empty by residue-count constraints, so $\tilde{e}_i^m(v_{k',e}) = 0$. $\square$

Example 10 ($e = 3, k' = 2$). At $(n, e, k') = (6, 3, 2)$ the target weight space at $m = 2$ requires $|\mu| = 4$ and residue count with a 0 coordinate. Direct enumeration of the 5 partitions of 4 gives residue counts

μ	(n_0, n_1, n_2)
(4)	(2, 1, 1)
(3, 1)	(1, 1, 2)
(2, 2)	(2, 1, 1)
(2, 1, 1)	(1, 2, 1)
(1, 1, 1, 1)	(2, 1, 1)

None equals $(0, 2, 2)$, $(2, 0, 2)$, or $(2, 2, 0)$; hence $\tilde{e}_i^2(v_{2,3}) = 0$ for every $i \in \{0, 1, 2\}$, matching the empirical WAKE 2026-08-10 result $\varepsilon_i(v_{2,3}) = 1$ for all i . Similarly at $m = 1, i = 0$: only $\mu = (3, 1, 1)$ has residue count $(1, 2, 2)$, so $\tilde{e}_0(v_{2,3})$ is a scalar multiple of $|(3, 1, 1)\rangle$ (in fact $\tilde{e}_0(v_{2,3}) = |(3, 1, 1)\rangle$).

7 Empirical observations beyond the theorem

The computational data at the four test triples (§8 below) reveals a pattern strictly stronger than the theorem: in every tested case, $\varepsilon_i(v_{k',e}) = 1$ for every i , well below the trivial bound $k' + 1$ from Theorem 5. We record the resulting conjecture, whose proof would subsume Theorems 5 and 8 in a much sharper form.

Conjecture 11 (Uniform depth-one). *For every $e \geq 2, k' \geq 1$ and every $i \in \{0, \dots, e-1\}$,*

$$\varepsilon_i(v_{k',e}) = 1.$$

Remark 12 (The conjecture is deeper than weight-space vanishing). *Proposition 9 settles $\varepsilon_i(v_{k',e}) \leq 1$ only when no partition of $ek' - 2$ has residue count $(k', \dots, k' - 2, \dots, k')$. But this hypothesis fails in general: at $(n, e, k') = (8, 2, 4)$ the 2-core $\kappa = (3, 2, 1)$ has size $6 = 2 \cdot 4 - 2$ and residue counts*

$(4, 2) = (k', k' - 2)$, so the weight space $\Lambda_0 - 4\delta + 2\alpha_1$ in $\hat{\mathfrak{sl}}_2$ -Fock is nonempty and Proposition 9 gives no bound. Nonetheless $\tilde{e}_1^2(v_{4,2}) = 0$ empirically.

Therefore Conjecture 11 cannot be proved by weight-space vanishing alone; it requires an actual cancellation in the linear combination $\tilde{e}_i^2 v_{k',e} = \sum \chi_{(e^{k'})}^\lambda \tilde{e}_i^2 |\lambda\rangle$. This cancellation is the interesting content: it says $v_{k',e}$, though not a single crystal basis element, still behaves as if it were a $\tilde{e}_i$ -depth-1 vector. The natural approach is via the boson–fermion correspondence $\mathcal{F} \cong V(\Lambda_0) \otimes \mathbb{Q}[p_e, p_{2e}, \dots]$, under which $v_{k',e} = |\emptyset\rangle_{V(\Lambda_0)} \otimes p_e^{k'}$; the Kashiwara operators $\tilde{e}_i$ do not commute cleanly with the Heisenberg multiplication at the crystal level (as evidenced by $\tilde{e}_i(v) \neq 0$), but the failure is controlled and depth-one.

The endpoint scalars along length- n balanced paths do *not* in general recover the character values $\chi_{(e^{k'})}^\lambda$:

- At $(6, 3, 2)$: distinct endpoint scalars $\{-2, -1, +1, +2\}$ equal the distinct values of $\chi_{(3,3)}^\lambda$. (Coincidence at small k' .)
- At $(9, 3, 3)$: distinct endpoint scalars $\{-6, -3, -1, +1, +3, +6\}$ vs. distinct values of $\chi_{(3,3,3)}^\lambda$ equal to $\{-3, -2, -1, +1, +2, +3, +6\}$. Not equal (endpoint has ± 6 ; character has ± 2).
- At $(12, 3, 4)$: distinct endpoint scalars $\{-18, -12, -8, -6, -4, -1, +1, +4, +6, +8, +10, +12, +16, +18\}$ vs. character values $\{-12, -8, -6, -4, -3, -1, +1, +2, +3, +4, +6, +8, +12\}$. The endpoint set has magnitudes up to 18, well above the character maximum 12.

So the endpoint-scalar $\leftrightarrow$ character-value correspondence conjectured in the PROVE seed (from the $(6, 3, 2)$ observation) does *not* generalise. The right formulation, if any, must involve weighted sums or refined statistics on the words. We leave this as an open question:

Conjecture 13 (Endpoint expansion, open). *For each balanced word w , let $\pi(w) \in \mathbb{Z}$ be the endpoint scalar $\tilde{e}_{w_n} \cdots \tilde{e}_{w_1}(v_{k',e}) = \pi(w) |\emptyset\rangle$. There is a combinatorial statistic $\varphi(w, \lambda)$ (perhaps depending on a canonical e -quotient-indexed decomposition of w) such that*

$$\pi(w) = \sum_{\lambda \vdash n} \varphi(w, \lambda) \chi_{(e^{k'})}^\lambda,$$

recovering the closed formula of PROVE 2026-08-08 as a specific sum.

8 Computational verification

The four test triples $(n, e, k') = (6, 3, 2), (8, 2, 4), (9, 3, 3), (12, 3, 4)$ were computed by ~/projects/probes/2026-08 Summary:

(n, e, k')	$ \text{supp}(v) $	residue count	ε_i (all i)	# nonzero balanced words
$(6, 3, 2)$	9	$(2, 2, 2)$	1	6 of 90
$(8, 2, 4)$	20	$(4, 4)$	1	1 of 70
$(9, 3, 3)$	22	$(3, 3, 3)$	1	17 of 1680
$(12, 3, 4)$	51	$(4, 4, 4)$	1	54 of 34650

In each case Theorem 4 (residue count $(k', \dots, k')$) and Theorem 6 (nonzero-path words are permutations of $(0^{k'} 1^{k'} \cdots (e-1)^{k'})$) are verified directly.

Raw verify.log

```
### (n, e, k') = (6, 3, 2)
|support of v| = 9
Distinct residue-count tuples in support: {(2, 2, 2)}
Weight-homogeneous of weight (k',...,k')? True
epsilon_i(v) for i=0..2: [1, 1, 1]
Theorem 1(b) bound k'+1 = 3; observed max epsilon = 1
Balanced words enumerated: 90 (elapsed 0.0s)
Nonzero endpoint scalars: 6
Multiset of nonzero endpoint scalars pi(w):
    pi = -2 occurs 1 times
    pi = -1 occurs 2 times
    pi = +1 occurs 2 times
    pi = +2 occurs 1 times
Distinct nonzero endpoint scalars: [-2, -1, 1, 2]
Distinct nonzero chi^lambda_(e^k') values: [-2, -1, 1, 2]

### (n, e, k') = (8, 2, 4)
|support of v| = 20
Distinct residue-count tuples in support: {(4, 4)}
Weight-homogeneous of weight (k',...,k')? True
epsilon_i(v) for i=0..1: [1, 1]
Theorem 1(b) bound k'+1 = 5; observed max epsilon = 1
Balanced words enumerated: 70 (elapsed 0.0s)
Nonzero endpoint scalars: 1
Multiset of nonzero endpoint scalars pi(w):
    pi = +1 occurs 1 times
Distinct nonzero endpoint scalars: [1]
Distinct nonzero chi^lambda_(e^k') values: [-6, -4, -3, -2, -1, 1, 3, 4, 6, 8]

### (n, e, k') = (9, 3, 3)
|support of v| = 22
Distinct residue-count tuples in support: {(3, 3, 3)}
Weight-homogeneous of weight (k',...,k')? True
epsilon_i(v) for i=0..2: [1, 1, 1]
Theorem 1(b) bound k'+1 = 4; observed max epsilon = 1
Balanced words enumerated: 1680 (elapsed 0.3s)
Nonzero endpoint scalars: 17
Multiset of nonzero endpoint scalars pi(w):
    pi = -6 occurs 1 times
    pi = -3 occurs 3 times
    pi = -1 occurs 3 times
    pi = +1 occurs 4 times
    pi = +3 occurs 5 times
    pi = +6 occurs 1 times
Distinct nonzero endpoint scalars: [-6, -3, -1, 1, 3, 6]
Distinct nonzero chi^lambda_(e^k') values: [-3, -2, -1, 1, 2, 3, 6]
```

```

### (n, e, k') = (12, 3, 4)
|support of v| = 51
Distinct residue-count tuples in support: {(4, 4, 4)}
Weight-homogeneous of weight (k', ..., k')? True
epsilon_i(v) for i=0..2: [1, 1, 1]
Theorem 1(b) bound k'+1 = 5; observed max epsilon = 1
Balanced words enumerated: 34650 (elapsed 110.1s)
Nonzero endpoint scalars: 54
Multiset of nonzero endpoint scalars pi(w):
    pi = -18 occurs 1 times
    pi = -12 occurs 1 times
    pi = -8 occurs 2 times
    pi = -6 occurs 3 times
    pi = -4 occurs 11 times
    pi = -1 occurs 4 times
    pi = +1 occurs 7 times
    pi = +4 occurs 13 times
    pi = +6 occurs 2 times
    pi = +8 occurs 5 times
    pi = +10 occurs 2 times
    pi = +12 occurs 1 times
    pi = +16 occurs 1 times
    pi = +18 occurs 1 times
Distinct nonzero endpoint scalars: [-18, -12, -8, -6, -4, -1, 1, 4, 6, 8, 10, 12, 16, 18]
Distinct nonzero chi^lambda_e^k' values: [-12, -8, -6, -4, -3, -1, 1, 2, 3, 4, 6, 8, 12]

```

9 Summary and post-PROVE targets

What is proved. Theorems 4, 5, and 6 in full generality (all $e \geq 2, k' \geq 1$). These are the (a), (b), and (d)-form claims of the PROVE seed.

What is corrected. Theorem 8 strictly refutes the seed's stretch conjecture (c): $\tilde{e}_0^{k'}(v_{k',e}) = 0$, not $\neq 0$. Proposition 9 gives the general weight-space vanishing framework in which (c) sits.

What is open (post-PROVE targets).

1. Conjecture 11: $\varepsilon_i(v_{k',e}) = 1$ for every i (empirically observed at $(6, 3, 2), (8, 2, 4), (9, 3, 3), (12, 3, 4)$; the upper bound reduces to a residue-count claim about e -cores; the lower bound requires a nontrivial noncancellation argument).
2. Conjecture 13: an explicit formula expressing the endpoint scalar $\pi(w)$ of each balanced word as a linear combination of character values $\chi_{(e^{k'})}^\lambda$ with combinatorial coefficients $\varphi(w, \lambda)$.
3. The crystal-theoretic reformulation of the closed formula proved on 2026-08-08: for each λ with $\chi_{(e^{k'})}^\lambda \neq 0$, does there exist a canonical balanced word $w(\lambda)$ (Yamanouchi with respect to the e -quotient) whose endpoint scalar equals $\chi_{(e^{k'})}^\lambda$? The $(6, 3, 2)$ data is consistent with

this (6 nonzero words vs. 9 support partitions with $\chi \neq 0$; multiplicities present); the larger cases suggest the correspondence is more delicate.

What this establishes for the §7 program. The three proved theorems are exactly the intrinsic $\widehat{\mathfrak{sl}}_e$ -crystal fingerprint of $v_{k',e}$: weight, Kashiwara-depth bound, and balanced-letter constraint on vacuum paths. Together they suffice to characterise $v_{k',e}$ as a Heisenberg-depth- k' vector in level-1 Fock. The endpoint-scalar $\leftrightarrow$ character-value bridge (Conjecture 13), which would upgrade this to a full crystal-theoretic proof of the 2026-08-08 closed formula, remains the natural next target.